

## Q.no 8:

### Question description

kkalaiselvan is going to behave as a vehicle driver.

he needs to drive an automobile on a track divided into "N" no. of sub-tracks.

The letter "K" is also assigned to you.

i.e. the maximum number of kilometres a vehicle may go on each sub-track.

You can add any unit of Petrol to the automobile if it can't cover a sub-track. The total kilometres your automobile may go increases by one unit for every unit of gasoline added.

Input:

The first line of input contains an integer T denoting the no of test cases. Then T test cases follow. Each test case contains two space separated integers N and K.

The second line of each test case contains N space separated integers (A[]) denoting the distance of each N sub-tracks.

Output:

For each test case in a new line you have to print out the minimum unit of Petrol your car require to cover all the sub-tracks. If no extra unit of petrol is required, print -1.

### ✓ Logical Test Cases

#### Test Case 1

INPUT (STDIN)

```
2
5 6
2 5 4 5 2
5 3
1 6 3 5 2
```

EXPECTED OUTPUT

```
-1
3
```

#### Test Case 2

INPUT (STDIN)

```
2
5 6
12 15 14 15 12
5 3
11 61 31 51 21
```

EXPECTED OUTPUT

```
9
58
```

### ✓ Mandatory Test Cases

Code:

```
#include <stdio.h>

void sort(int a[], int n)
{
    int i, j, temp;
```

```

    for (i = 0; i < n - 1; i++)
    {
        for (j = i + 1; j < n; j++)
        {
            if (a[i] > a[j])
            {
                temp = a[i];
                a[i] = a[j];
                a[j] = temp;
            }
        }
    }
}

int main()
{
    int T;
    scanf("%d", &T);

    int answer[100];

    for (int t = 0; t < T; t++)
    {
        int N, K;
        int A[100];

        scanf("%d %d", &N, &K);

        for (int i = 0; i < N; i++)
        {
            scanf("%d", &A[i]);
        }

        sort(A, N);

        if (A[N - 1] <= K)
        {
            answer[t] = -1;
        }
        else
        {
            answer[t] = A[N - 1] - K;
        }
    }

    printf("Output:\n");

```

```
for (int t = 0; t < T; t++)  
{  
    printf("%d\n", answer[t]);  
}  
  
return 0;  
}
```

Output:

Test1:

```
addityajaiswal@Addityas-MacBook-Air:  
2  
5 6  
2 5 4 5 2  
5 3  
1 6 3 5 2  
Output:  
-1  
3  
addityajaiswal@Addityas-MacBook-Air:
```

Test 2:

```
addityajaiswal@Addityas-MacBook-Air:  
2  
5 6  
12 15 14 15 12  
5 3  
11 61 31 51 21  
Output:  
9  
58  
addityajaiswal@Addityas-MacBook-Air:
```